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EMOTION REGULATION AND MENTAL HEALTH OUTCOMES BETWEEN SOCIOECONOMIC CLASSES AMONG UNIVERSITY STUDENTS

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Abstract – This study investigated emotion regulation and its impact on mental health. This modified replication of Troy et al. (2017), showed that socioeconomic status (SES) moderated the relationship between cognitive reappraisal ability (CRA) and depression while ruling out life stress. Our first hypothesis replicated the original study, the second predicted the interaction effect of CRA and SES on satisfaction with life (SWL), and the third examined whether SES moderates emotional suppression ability (ESA) and depression. Undergraduate students (N = 118) completed an online survey. Our findings were inconsistent with those of Troy et al. (2017). SES did not moderate between CRA and depression, CRA and SWL, or ESA and depression. The results indicated the benefits of emotion regulation on mental health may apply regardless of their SES.

Keywords: emotional regulation, cognitive reappraisal ability, socioeconomic status, mental health, emotion suppression, depression

Every human being experiences a vast multitude of differing emotions throughout each day, but how one chooses to express or suppress these emotions may have a larger impact on our mental well-being than some may think. Some individuals are inherently more effective in regulating their emotions toward specific situations. Specifically, how an individual chooses to regulate their emotions and interpret their thoughts can affect negative mental health outcomes, such as depression and personal dissatisfaction (Troy et al., 2013). Our research represents a tailored replication of an investigation conducted by Troy et al. (2017), which delved into the intricate relationship between cognitive reappraisal ability (CRA), socioeconomic status (SES), life stress, and symptoms of depression. Building upon the foundations of the Troy et al. (2017) study, our research seeks to contribute to the understanding of how SES might moderate the associations between three sets of variables: CRA and depression, CRA and satisfaction with life (SWL), and emotion suppression ability (ESA) and depression.

The effect of CRA and SES on depression (Hypothesis 1: direct replication)

Troy et al. (2017) examined the interplay between CRA, SES, and depression. SES can be defined as a person's relative social standing determined by their household income, educational achievements, and overall assets (Kraus et al., 2012). A person's ownership over these assets can determine their ability to control their external environment and overall satisfaction. Individuals with lower SES experience heightened stress but possess fewer coping resources, leading to decreased psychological well-being (Finkelstein et al., 2007; Navarro-Carrillo et al., 2020). Consequently, they often feel less control over their lives with their limited resources (Gallo et al., 2005; Kraus et al., 2009). Those who lack these benefits are subject to harsher financial circumstances, having to rely on mental fortitude and emotional revaluation to bring happiness into their lives (Kraus et al., 2012).

As well as this, CRA, defined as the capacity to reframe emotional situations to alter their emotional impact (Goldin et al., 2012), was examined in relation to SES and its impact on a person's negative thoughts and feelings toward themselves or their environment otherwise known as depression (Troy et al., 2017). The findings revealed a moderation effect which confirmed their hypothesis, suggesting that higher CRA was significantly associated with having lower depressive symptoms in individuals with lower SES, although the same relationship was notably weaker in individuals with higher SES after controlling for life stress. Furthermore, cognitive reappraisal is a dependable emotion regulation strategy, particularly helpful in uncontrollable situations (Roos & Bennett, 2022). According to Brandao et al.

(2023), individuals who struggle to regulate their emotions are at greater risk of developing depressive symptoms. This indicates that emotion regulation can be critical for mental health, especially when individuals lack sufficient material resources to control their situations.

Especially in cases of low SES where individuals lack control over their life trajectory (Gallo et al., 2005; Kraus et al., 2009), CRA could play a critical role in determining mental health, whereas it may have less of an effect on high SES individuals. Research conducted by Cohen et al. (1983) contributes to our understanding of life stress as a potential confounder in studies exploring the relationship between emotion regulation and psychological health issues like depression. Based on these findings, we aim to expand the research and increase generalizability through direct and modified replication. Our first hypothesis is a direct replication of Troy et al. (2017), investigating if there is truly an interaction between an individual's CRA and SES on depressive symptoms after controlling for life stress. By controlling for life stress, we aim to build on insights from Cohen et al. (1983) and isolate the specific impacts of SES and emotion regulation techniques on mental health.

The effect of CRA and SES on Life Satisfaction (Hypothesis 2: expansion)

In our second hypothesis, our focus shifted to whether SES moderates the link between CRA and SWL while ruling out the potentially confounding effects of life stress. In this context, SWL is an individual's cognitive evaluation of their own life, measuring whether they feel that their life is meaningful and holds significant value (Mehrotra et al., 2018). Taking this into account, CRA in specific contexts is also linked to positive psychological effects, such as higher SWL, a greater sense of self-esteem, and improved overall well-being (Gross, 1998; Troy et al., 2013). Also, SES can be related to SWL: A meta-analysis showed that individuals with high SES were more likely to be satisfied with life (Moreno-Agostino et al., 2021). High SES individuals are less concerned about their finances, which leads to an increase in life satisfaction, while those with lower SES are faced with financial difficulties and are more dissatisfied with their circumstances and lives (Ansah et al., 2022). Since high SES individuals have more control over their lives (Gallo et al., 2005; Kraus et al., 2009), they do not necessarily need to have the same level of control over their emotions, so the ability to control their emotions may not be related to their SWL in the same way it would be for those of low SES (Heidemeier, 2017). Therefore, we expand the findings of Troy et al. (2017), and predict the interaction between CRA and SES on SWL: upper-class individuals will not have a relationship between CRA and SWL, while lower-class individuals will have a positive relationship between the two.

The effect of ESA and SES on Depression (Hypothesis 3: expansion)

In our third hypothesis, we seek to modify and extend Troy et al. (2017) by focusing on ESA instead of CRA, while ruling out the confounding effects of life stress. We aim to shed light on a different facet of emotion regulation (i.e., ESA), as well as contribute valuable insights into how various emotion regulation strategies might interact with SES to influence mental health outcomes. In this regard, ESA refers to an individual's capacity to consciously inhibit or restrain the expression of their emotions (Gross, 1998). In other words, ESA is the capability of an individual to suppress their emotions and feelings when necessary, not the frequency with which they are suppressed. Gross and John (2003) provide valuable insights into the mechanisms of emotion regulation. Their Emotion Regulation Questionnaire (ERQ), which includes subscales for both cognitive reappraisal and suppression, provides a foundation for investigating ESA. Gross and John (2003) demonstrated that individuals who frequently employ emotion suppression strategies may experience negative psychological consequences, such as increased stress and depressive symptoms. While their findings demonstrate the role that suppression frequency plays in health outcomes, our interest lies in the effects of suppression ability. Snibbe and Markus (2005) explored how educational attainment, closely related to SES, affects agency and choice in individuals. They found that higher SES individuals tend to perceive themselves as having more control over their life choices and situations, whereas lower SES individuals often feel more constrained.

Based on these findings, we expect the strength of ESA's effect on depressive severity to be moderated by each individual's SES. While the act of suppressing emotions can either help or hurt depending on the context, we posit that the ability to suppress and control one's emotions is a universally beneficial skill (Bonanno et al., 2004; Burton et al., 2016). However, that skill may be more beneficial for those who have relatively less agency and control over their standing in the socioeconomic landscape, as they often cannot afford to let their emotions get the best of them. For high SES individuals, we hypothesize that ESA and depression will have a weak negative relationship, but for low SES individuals, we hypothesize a strong negative relationship.

Overview of Current Research

Our study seeks to understand how individuals from different socioeconomic backgrounds navigate the challenges of emotion regulation strategies and their implications. The sample for our study consists of 118 university students at University of Central Arkansas (UCA). Based on the literature, we expect that SES may exert varying influences on the relationships between our variables, with the strength of these relationships contingent upon the individual's SES. This study is part of the Collaborative Replication and Education Project (CREP) and was conducted for a graded class assignment at UCA. All research data and materials are shared in Open Science Framework (https://osf.io/cu6p4/). The replication plan and hypothesis were registered before data collection (https://osf.io/t6hfb).

Method

Participants

A total of 118 participants were recruited for this study. As part of CREP, the sample size had been determined to be more than 100 university participants for each university in advance. Those who were under the minimum age of 18, who finished the study in less than 3 minutes, or who did not fully complete the survey were excluded from the final data analysis as pre-registered. Participants were adults with ages ranging from 18-47 (M = 21.24, SD = 4.50). Participants were predominantly women, with 83.9% identifying as female, 12.7% identifying as male, 2.5% identifying as non-binary, and 0.8% preferred not to say. The participants' racial identity consisted of 76.3% being Caucasian, 12.7% being African American, 5.1% being multiple races, 3.4% identifying as other, 1.7% being Asian, and 0.8% preferred not to say. Their ethnic background included 84.7% identifying as not Hispanic/Latino, 6.8% identifying that they were Hispanic/Latino, and 4.2% preferring not to say.

Procedure

This current study was approved by the Institutional Review Board through the authors' institution, as we have replicated the study conducted by Troy et al., (2017). Participants were recruited through the university's SONA system, through convenience sampling, and were provided online consent to participate in this study. Participants completed an online survey that took approximately 15 minutes to complete, which started with demographic questions asking for gender, age, racial/ethnic identity, ethnicity, student status, and current annual family income (SES). After answering the demographic questions, participants answered questions regarding CRA, habitual cognitive reappraisal use, depressive symptoms, and life stress using the original scales from Troy et al. (2017), along with the additional scales of SWL, ESA, and habitual emotion suppression use (HESU) from the current study. Upon completion of the online survey, participants were thanked, debriefed, and compensated for their time with SONA credits.

Measures

Socioeconomic Status (SES)

We measured SES at the beginning of the survey via the demographic questions. We used the exact question and scale to replicate Troy et al. (2017). We asked participants what their family income was with this question "How high is your current annual family income?", rated on a 1 ("$10,000 or less") to 12 ("$200,000 or above") scale (M = 21.24, SD = 3.50), as a measurement for SES (Table 1). Many individuals find it hard to determine and report their exact annual family income level, therefore we asked them to score it using a series of estimated ranges, as was done by Troy et al. (2017).

Table 1

Distribution of Annual Family Income

Annual Family Income	% of Total
$10,000 or less	9.3 %
$10,001-$20,000	6.8 %
$20,001-$30,000	7.6 %
$30,001-$40,000	10.2 %
$40,001-$50,000	7.6 %
$50,001-$60,000	10.2 %
$60,001-$70,000	7.6 %
$70,001-$80,000	4.2 %
$80,001-$90,000	5.1 %
$90,001-$100,000	7.6 %
$100,001-$200,000	17.8 %
$200,000 or more	5.9 %

Cognitive Reappraisal Ability (CRA)

We measured CRA with an 8-item measure that was adapted from the Emotion Regulation Questionnaire created by Gross and John (2003), as used in Troy et al. (2017). The questions were modified to measure an

individual's ability to use cognitive reappraisal with strong reliability (α = .90). Participants were asked to rate the questions on a scale of 1 (*strongly disagree*) to 7 (*strongly agree*), with some examples including: "When I really want to, I am very capable of changing what I'm thinking about when I want to feel less negative emotion (such as sadness or anger)," and "When I really want to, I am very capable of reconsidering what relevance the situation really has for me if a situation is likely to make me upset." Higher scores indicated a greater ability to reappraise one's cognitions about emotions, and the total CRA score was measured by calculating the mean of all 8 questions (M = 4.78, SD = 1.14).

Habitual Cognitive Reappraisal Use (HCRU)

Habitual cognitive reappraisal use was measured using a subscale of the ERQ (Gross & John, 2003), and consists of six questions which are answered on a scale of 1 (*strongly disagree*) to 7 (*strongly agree*), as used in Troy et al. (2017). Some questions asked were: 'When I want to feel less negative emotion (such as sadness or anger), I change what I'm thinking about," and "I control my emotions by changing the way I think about the situation I'm in." Participants with higher scores in this category indicated they were more likely to use cognitive reappraisal as an emotion regulation strategy. The total score was measured for each person by calculating the mean of all six questions (M = 4.84, SD = 1.09, α = .89).

Depressive Symptoms

Depressive symptoms were measured using a 5-item questionnaire adapted from the Center for Epidemiologic Studies Depression Scale (Radloff, 1977). Participants were asked to report symptoms of depression that they noticed within the past week, on a scale of 0 (*rarely or none of the time*) to 3 (*most of the time*). Examples of symptoms inquired in this section of the survey include: "I felt depressed," and "I had crying spells." Higher scores indicated a greater presence of depressive symptoms, and the total score was calculated by taking the sum of all 5 questions (M = 10.86, SD = 3.63, α = .75).

Life Stress

Life stress was measured using a 4-item questionnaire that was adapted from the Perceived Stress Scale (Cohen et al., 1983). Participants were asked to report their perceptions of stress over the past two years, with items being rated on a scale of 1 (*never*) to 5 (*very often*). Some examples of questions included are: "In the past two years, how often have you felt that things were going your way? (R)" and "In the past two years, how often have you felt that you were unable to control the important things in your life?" Higher scores indicated a greater presence of life stress in individuals, and the total score was assessed by taking the mean of all 4 questions (M = 3.04, SD = 0.71, α = .74).

Satisfaction with Life (SWL)

We measured SWL using the 5-item SWL scale questionnaire created by Diener et al. (1985) with participants being asked how content they are with life, on a scale of 1 (*strongly disagree*) and 7 (*strongly agree*). Some examples of questions included are: "In most ways, my life is close to ideal," and "If I could live my life over, I would change almost nothing." Higher scores of SWL indicated that the individual was overall more satisfied with the circumstances of their life, and the total score was then assessed by taking the mean of all 5 questions (M = 4.09, SD = 1.23, α = .85).

Emotion Suppression Ability (ESA)

We measured ESA by using a 4-item scale, adapted from Gross and John's (2003) ERQ, with participants being asked to report their ability to suppress emotions on a scale of 1 to 7 (1 being *strongly disagree* and 7 being *strongly agree*). We adjusted the items similarly to CRA to measure the emotion suppression ability, rather than the frequency of their use. Some examples of questions included are: "When I really want to, I am capable of keeping emotions to myself," and "When I really want to, I am capable of controlling my emotions by not expressing them." Higher scores indicated a greater ability to suppress one's emotions when necessary, and the total score was assessed by taking the mean of all 4 questions (M = 4.96, SD = 1.26, α = .81).

Habitual Emotion Suppression Use (HESU)

We measured HESU using a questionnaire taken directly from Gross and John's (2003) ERQ and consisted of 4 questions. Participants were asked to report their HESU on a scale of 1 (*strongly disagree*) to 7 (*strongly agree*). Example questions include: "I control my emotions by not expressing them," and "I keep my emotions to myself." Higher scores indicated more frequent or habitual use of emotion suppression in the surveyed individuals, and the total score was assessed by taking the mean of all 4 questions (M = 3.89, SD = 1.33, α = .78).

Table 2
Correlations for Study Variables

Variable	1	2	3	4	5	6	7
1. CRA	—						
2. ESA	.13	—					
3. SES	.15	.10	—				
4. Depressive symptoms	-.34***	-.07	-.10	—			
5. SWLS	.46***	-.01	.09	-.49***	—		
6. HCRU	.74***	.09	-.03	-.33***	.44***	—	
7. HESU	-.13	.52***	-.11	.17	-.20*	-.07	—
8. Life stress	-.39***	-.00	-.12	.53***	-.56***	-.36***	.20*

*p < .05. **p < .01. ***p < .001

Results

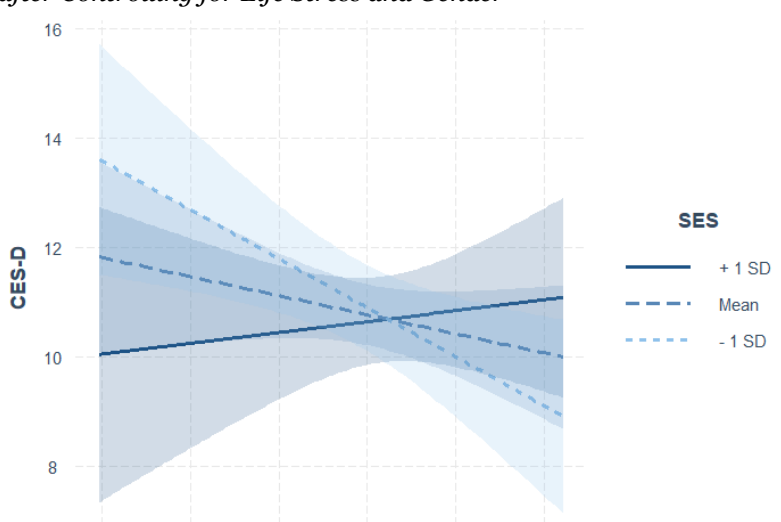
As a first analysis, we examined the associations between all of the measures in the study. The results of these Pearson's correlations are reported in Table 2.

For the interaction analyses, the continuous variables in the regression models were mean-centered. To control for other variables, we included them as predictors in the regression models.

To examine hypothesis 1, we conducted a multiple regression to test the interaction between SES and CRA on depression, which was not significant after controlling for life stress, β = 0.12, t = 1.64, p = .104, 95% CI = [-0.02, 0.26]. These findings were inconsistent with Troy et al. (2017), as SES did not moderate the relationship between CRA and depression. That said, when we examined the simple slopes by SES, the patterns were consistent with the hypothesis. For those with higher SES, simple slope analyses indicated that CRA was not associated with depressive symptoms, β = -0.02, t = -0.04, p = .96. However, for those lower in SES, higher CRA was negatively associated with depression, β = -0.84, t = -2.45, p = .02. For lower SES participants, when they perceived that they are good at cognitive reappraisal, they reported less depression.

We added demographic variables (age, sex, and race), and habitual reappraisal use individually as control variables to the model described above by Troy et al. (2017). Gender was dummy-coded (1: women, 0: men and other options). When we controlled for life stress and gender, the interaction became significant (Figure 1), β = 0.15, t = 2.12, p = .036, 95% CI = [0.01, 0.30]. As predicted by the hypothesis, for lower SES, higher CRA was significantly related to lower depression, β = -1.05, t = -2.77, p = .007, 95% CI = [-1.81, -0.30]. And it was not significant for higher SES, β = 0.31, t = 0.72, p = .471, 95% CI = [-0.55, 1.18]. Also, when life stress and HCRU were controlled for, the interaction was marginally significant, β = 0.12, t = 1.76, p = .081, 95% CI = [-0.02, 0.27]. For lower SES, CRA was not related to depression, β = -0.68, t = -1.48, p = .143, 95% CI = [-1.60, 0.23]. And it was not significant for higher SES, β = 0.43, t = 0.80, p = .425, 95% CI = [-

Figure 1
The Interaction between CRA and SES on Depressive Symptoms after Controlling for Life Stress and Gender



0.64, 1.51]. In other models, the interaction between CRA and SES was not significant.

When assessing hypothesis 2, we conducted a multiple regression to evaluate the interaction between CRA and SES on SWL, which was not significant after controlling for life stress, $\beta = -0.01$, $t = -0.24$, $p = .812$, CI = [-0.05, 0.04]. However, the main effect of CRA was significant, which indicates that higher CRA was related to higher satisfaction with life, $\beta = 0.31$, $t = 3.50$, $p < .001$, CI = [0.13, 0.48]. That being said, SES was not found to be a moderator between CRA and SWL, and there was a positive association between the two variables regardless of one's SES. When we examined the effect of demographics or HCRU individually, the interactions between CRA and SES were still not significant.

To examine hypothesis 3, we conducted a multiple regression to test the interaction between SES and ESA on depression and found that it was not significant after controlling for life stress, $\beta = 0.11$, $t = 1.55$, $p = .124$, CI = [-0.03, 0.25]. For individuals with higher SES, simple slope analyses indicated that ESA was not associated with depressive symptoms, $\beta = 0.23$, $t = 0.65$, $p = .52$. For individuals with lower SES, higher ESA was marginally associated with having fewer depressive symptoms, $\beta = -0.54$, $t = -1.70$, $p = .09$. Therefore, SES did not moderate the relationship between ESA and depressive symptoms. When we tested the effect of demographics or HESU individually, the interactions between ESA and SES were still not significant.

Discussion

Our goal for this present study is to measure the moderating effects of SES on the link between CRA and depression, CRA and SWL, and ESA and depression while ruling out the confounding effects of life stress. For each relationship, we predicted that SES would be a moderator between the variables. However, the results of this study did not support our hypotheses as we found that SES was not a moderating factor between any of the variables used.

Implications of our Study

In general, our study result did not replicate the result of Troy et al. (2017). That said, we discovered some evidence supporting the direct replication hypothesis. There was the significant interaction between CRA and SES on depression as expected when controlled for life stress and gender. Specifically, when controlling for life stress and gender, we found that individuals who perceived themselves to be effective at changing their emotions reported lower levels of depression, but this effect was primarily observed in lower SES participants. This suggests that the ability to regulate emotions through CRA may play a crucial role in mitigating depressive symptoms, particularly for those facing socioeconomic difficulties, as consistent with previous research (Chen & Miller, 2012). Also, given the research about the gender differences in the emotion regulation ability (e.g., Min et al., 2023), controlling for gender might have been able to show the predicted relationship more clearly. Additionally, we found the marginally significant interaction between CRA and SES on depression when controlled for life stress and HCRU.

Furthermore, we found that higher CRA was associated with fewer depressive symptoms, indicating that those who feel more capable of employing cognitive reappraisal strategies experience fewer symptoms of depression. These findings are consistent with previous studies, showing that better emotion regulation strategies can help overall mental well-being (Roos & Bennet, 2022). We also observed a correlation between higher CRA and greater SWL across all SES groups, suggesting that effective emotional regulation may enhance overall life satisfaction, regardless of socioeconomic circumstances. That being said, Troy et al. (2013) and Gross (1998), also found that CRA is linked to positive psychological effects, one being SWL. Interestingly, we found no relationship between ESA and depression, indicating that the mechanisms underlying these variables may differ.

Strengths of our Study

Our study provides valuable insights into the role of CRA in the context of SES and depression. Our findings suggest that higher CRA is associated with reduced depressive symptoms across different socioeconomic backgrounds. Future studies should aim to replicate these results in more diverse and representative samples to better understand the relationship between emotion regulation strategies, SES, and mental health. Through our systematic replication and adaptation of the Troy et al. (2017) research, we hope to enrich the existing literature on emotion regulation by offering a comprehensive view of how SES and emotion regulation strategies influence depressive symptoms and satisfaction with life. Our findings carry the potential to illuminate the varying experiences of individuals from diverse socioeconomic backgrounds and their strategies for managing emotional distress. Ultimately, we hope that the knowledge generated from this study can inform interventions and support systems tailored to the unique needs of different socioeconomic groups, promoting mental health and well-being for all.

Limitations and Future Research Directions

The discrepancies between our findings and those of Troy et al. (2017) highlight the influence of participant demographics. The inconsistent findings

could be due to the focus on female participants in Study 2 in Troy et al. (2017), which can alter the significant interaction between CRA and SES when not taking the male population into account. As well as this, Troy et al. (2017) study did not focus on college students which can lead to contrasting results as college students have more stressors in life while juggling school work and jobs.

One of the limitations of our research was the sample size of the population tested, leading to a lack of variation, which consisted entirely of students from a university in the West South Central region of the U.S., with the vast majority identifying as female (83.9%), severely limiting the sample's scope and diversity. This lack of diversity in our population limits the generalizability of our findings and raises concerns about whether the results can be applied to other populations or contexts. Future research would benefit from incorporating a more diverse population sample inclusive of gender and ethnicity. Also, the cross-sectional design of our study, meaning that all of the data was collected at one point in time rather than longitudinally collecting data repeatedly over a longer period, may have restricted our ability to draw conclusions regarding the relationships between variables.

In addition, the SES measurement we used from Troy et al. (2017), had individuals report their average annual family income which was limited to a series of broad categories, all of which would have to be mentally estimated and selected on the spot with little time to prepare a thoughtful response. Notably, out of the 12 SES categories available for selection, almost 1/5 of our sample (17.8%) selected the "$100,001 - $200,000" response. This is an incredibly broad estimate, with a range of nearly one hundred thousand dollars, which could hinder our analysis concerning SES moderation. It may benefit future researchers to include a more thorough measure of SES among participants, potentially incorporating past and present opportunities for education, housing, employment, and social relationships or organizations (Antonoplis, 2023). Moreover, examining other potential moderators, such as cultural or religious factors (Chesser et al., 2018; Nam et al., 2018), can contribute to a more comprehensive understanding of these relationships.

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